

## CAPE FLATTERY LH, WA, USA

WMO: 994300

Lat: 48.3915N

Lon: 124.7370W

Elev: 31

StdP: 100.96

Time Zone: -8.00 (NAP)

Period: 94-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 0.0        | 1.6 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 26.9                  | 8.3  | 23.7 | 6.4  | 9.9                   | 90   | 0.754 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 3.3                    | 17.6            | N/A  | 16.5 | N/A  | 15.6 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 4.4                  | 80   |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               | Max  |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 21.8              | 18.2 | 15.5 | -1.5                       | 22.2 | 2.0                | 2.0 | -3.0                                               | 23.6 | -4.2       | 24.8 | -5.3       | 25.9 | -6.8       | 27.4 |
|                   |      |      | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.7    | 6.6   | 6.6   | 7.2   | 8.4   | 10.2  | 11.9  | 13.0  | 13.3  | 12.8  | 10.9  | 8.2   | 6.6  |
|                                                                                  | DBStd       | 3.18   | 2.43  | 2.17  | 1.80  | 1.72  | 1.68  | 1.50  | 1.56  | 1.30  | 1.56  | 1.70  | 2.26  | 2.32 |
|                                                                                  | HDD10.0     | 542    | 107   | 97    | 88    | 53    | 16    | 2     | 0     | 0     | 1     | 9     | 61    | 108  |
|                                                                                  | HDD18.3     | 3164   | 364   | 329   | 344   | 298   | 251   | 193   | 164   | 156   | 167   | 230   | 303   | 365  |
|                                                                                  | CDD10.0     | 417    | 1     | 1     | 2     | 6     | 23    | 59    | 94    | 103   | 83    | 38    | 7     | 2    |
|                                                                                  | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 6.3    | 8.8   | 8.3   | 6.9   | 5.9   | 5.2   | 4.3   | 4.5   | 4.5   | 4.8   | 6.3   | 7.7   | 8.1  |
| Precipitation                                                                    | PrecAvg     | 2639   | 441   | 259   | 265   | 187   | 100   | 75    | 49    | 71    | 96    | 276   | 425   | 398  |
|                                                                                  | PrecMax     | 3471   | 694   | 674   | 512   | 364   | 180   | 168   | 116   | 310   | 277   | 547   | 763   | 654  |
|                                                                                  | PrecMin     | 2032   | 260   | 31    | 39    | 47    | 19    | 25    | 5     | 5     | 10    | 31    | 168   | 199  |
|                                                                                  | PrecStd     | 385    | 117   | 148   | 117   | 88    | 49    | 38    | 32    | 74    | 63    | 122   | 180   | 105  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.3  | 12.2  | 12.9  | 14.7  | 17.5  | 19.5  | 19.9  | 18.7  | 18.5  | 16.6  | 14.0  | 12.7 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | DB     | 10.9  | 11.0  | 11.3  | 12.6  | 14.3  | 15.8  | 17.0  | 16.9  | 16.8  | 15.1  | 12.8  | 11.3 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | DB     | 10.3  | 10.1  | 10.4  | 11.4  | 13.0  | 14.6  | 15.9  | 16.0  | 15.8  | 14.1  | 12.0  | 10.3 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | DB     | 9.6   | 9.3   | 9.7   | 10.5  | 12.3  | 13.8  | 15.2  | 15.3  | 15.1  | 13.4  | 11.2  | 9.5  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 2.7   | 2.8   | 3.1   | 3.1   | 3.0   | 3.1   | 3.1   | 3.3   | 3.7   | 3.1   | 2.8   | 2.7  |
|                                                                                  |             | MCDBR  | 3.5   | 3.5   | 4.0   | 4.3   | 4.3   | 4.4   | 4.6   | 4.7   | 5.0   | 3.8   | 3.5   | 3.5  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.317  | 0.314 | 0.333 | 0.355 | 0.372 | 0.354 | 0.346 | 0.348 | 0.339 | 0.333 | 0.319 | 0.311 |      |
|                                                                                  | taud        | 2.525  | 2.571 | 2.527 | 2.455 | 2.424 | 2.495 | 2.560 | 2.553 | 2.569 | 2.561 | 2.537 | 2.507 |      |
|                                                                                  | Ebn at Noon | 762    | 857   | 889   | 897   | 890   | 907   | 910   | 895   | 872   | 818   | 751   | 718   |      |
|                                                                                  | Edh at Noon | 61     | 72    | 87    | 103   | 110   | 104   | 96    | 92    | 83    | 72    | 60    | 55    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.92   | 1.81  | 2.75  | 4.21  | 5.24  | 5.73  | 5.89  | 4.93  | 3.80  | 2.17  | 1.11  | 0.76  |      |
|                                                                                  | RadStd      | 0.13   | 0.25  | 0.32  | 0.38  | 0.41  | 0.50  | 0.36  | 0.40  | 0.29  | 0.28  | 0.18  | 0.10  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (62 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page